

		Thus, $u_1 = 1.87 \times 10^7 \text{ m s}^{-1}$
5	D	From the diagram, it is obvious that the net force is not zero (all forces are pointing upwards or have a vertical component pointing upwards) Taking the centre of the disc as a pivot, the sum of anti-clockwise moments = $(15 + 30)r$, where r is the radius of the disc. This expression is equal to the sum of clockwise moments = $(20 + 25)r$.
6	A	Net work done = change in KE